



QUANTITATIVE METHODS 1 · CHAPTER 6

Statistical Index Numbers

Detailed solutions

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1. Elementary index and rebasing

A service price rises from EUR 40 in 2023 to EUR 46 in 2025.

- 1 Calculate the 2025 index with 2023=100 and the percentage change.
- 2 An old series is 90, 100, 108 for 2022-2024 with 2023=100. Rebase it to 2022=100.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

Detailed solution

Result. $I = 46/40 \times 100 = 115$, so the price rose by 15%. For rebasing, divide by 90 and multiply by 100: 100.00, 111.11, 120.00. Interpretation. Rebasing changes the reference, not the underlying relative movements.

2. Value, Laspeyres and Paasche

Product	p_0	q_0	p_1	q_1
A	2	100	2.5	90
B	5	40	6	50

Calculate the value index and Laspeyres and Paasche price indices.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

Detailed solution

Method. Compute $\sum p_0q_0=400$, $\sum p_1q_1=525$, $\sum p_1q_0=490$ and $\sum p_0q_1=430$.

Result. $I_V=525/400 \times 100=131.25$; $I_P^L=490/400 \times 100=122.50$;

$I_P^P=525/430 \times 100=122.09$. Interpretation. Value rose 31.25%, while the two price measures indicate about 22% inflation. They differ because their quantity weights come from different periods.

3. Fisher and factor reversal

Using Exercise 2, calculate Fisher's price index, Laspeyres and Paasche quantity indices, Fisher's quantity index and verify factor reversal.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

Detailed solution

Result. $I_P^F = \sqrt{122.50 \times 122.09} = 122.30$. $I_Q^L = 430/400 \times 100 = 107.50$ and $I_Q^P = 525/490 \times 100 = 107.14$, so $I_Q^F = 107.32$. Then $122.30 \times 107.32 / 100 = 131.25$ up to rounding, equal to I_V . Interpretation. Fisher provides a symmetric decomposition of value change into price and quantity components.

4. Fixed-base, chain and deflation

An index is 100 in year 0 and 110 in year 1. The year-2 link index relative to year 1 is 106. Nominal year-2 revenue is EUR 636,000.

- 1 Construct the chain index for year 2.
- 2 Deflate revenue into year-0 euros.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

Detailed solution

Result. Chain index $= 110 \times 106 / 100 = 116.6$. Real revenue $= 636,000 / (116.6 / 100) = 545,455$ euros approximately. Interpretation. Nominal revenue must be divided by the price-index factor, not merely reduced by 16.6%.

5. Supplier dashboard decision

A buyer reports: value index 128, Fisher price index 120 and Fisher quantity index 106.67. Explain the three numbers, verify consistency, and give two cautions before attributing the price rise to supplier market power.

Method prompt. Identify the statistical concept, state the relevant formula or rule, calculate carefully, and finish with an interpretation in context.

Detailed solution

Result. Expenditure is 28% above base, prices about 20% above base and quantities about 6.67% above base. Consistency holds because $120 \times 106.67 / 100 = 128$.

Interpretation. The index is descriptive. Before claiming market power, check product-quality and mix changes, contract terms, input costs, exchange rates, basket coverage and whether weights remain representative.

6. Complete price-quantity decomposition

A purchasing department observes:

Product	p0	q0	p1	q1
A	4	100	5	90
B	10	50	11	60
C	20	20	18	30

- 1 Calculate the four totals Σp_0q_0 , Σp_1q_0 , Σp_0q_1 and Σp_1q_1 .
- 2 Calculate the value, Laspeyres-price, Paasche-price and Fisher-price indices.
- 3 Interpret the gap between value and price change.

Method prompt. Complete all four cross-value columns before forming any index.

Detailed solution

Four totals. $\Sigma p_0q_0 = 400 + 500 + 400 = 1,300$. $\Sigma p_1q_0 = 500 + 550 + 360 = 1,410$.
 $\Sigma p_0q_1 = 360 + 600 + 600 = 1,560$. $\Sigma p_1q_1 = 450 + 660 + 540 = 1,650$.

Value index. $1,650 / 1,300 \times 100 = 126.92$.

Laspeyres price. $1,410 / 1,300 \times 100 = 108.46$. Paasche price. $1,650 / 1,560 \times 100 = 105.77$. Fisher price. $\sqrt{(108.46 \times 105.77)} = 107.11$.

Interpretation. Total purchasing value increased by 26.92%, whereas the balanced price increase is about 7.11%. Most of the value rise therefore reflects a positive quantity or mix effect, not prices alone. The formulas do not by themselves establish why quantities changed.

7. Rebasing, inflation and real performance

A nominal revenue series is €500k, €560k and €630k. The corresponding price index (year 1 = 100) is 100, 108 and 120.

- 1 Calculate real revenue in year-1 euros.
- 2 Calculate nominal and real growth from year 1 to year 3.
- 3 Rebase the price index so year 2 = 100.
- 4 Explain the result to a manager who claims that activity grew by 26%.

Method prompt. Deflate by dividing nominal value by index/100. Rebasing changes the reference, not the underlying inflation path.

Detailed solution

Real revenue. Year 1: $500/1 = €500k$. Year 2: $560/1.08 = €518.52k$. Year 3: $630/1.20 = €525k$.

Growth. Nominal growth is $630/500 - 1 = 26\%$. Real growth is $525/500 - 1 = 5\%$.

Rebased price index. Divide each value by 108 and multiply by 100: 92.59, 100, 111.11.

Managerial interpretation. Revenue received in current euros rose by 26%, but after adjusting for the 20% rise in the price level, purchasing-power-equivalent revenue rose by only 5%. Rebasing to year 2 changes the displayed levels, not this economic conclusion. The selected deflator must match the firm's product mix and coverage.